higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
NQF LEVEL 4

13 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 1.1.1 B
1.1.2 C
1.1.3 A
1.1.4 C
1.1.5 B

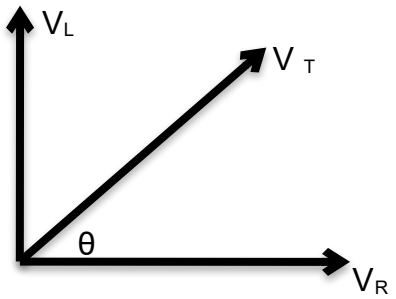
(5 × 1) (5)

1.2 1.2.1 $V_T = \sqrt{V_R{}^2 + V_L{}^2}$✓
$= \sqrt{30^2 + 40^2}$✓
$= \sqrt{900 + 1\,600}$
$= 50\ V$✓ (3)

1.2.2 $\Theta = \cos^{-1} V_R/V_T$✓
$= \cos^{-1} 30/50$✓
$= 53{,}13°$✓ (3)

1.2.3 $Z = V_T/I_T$✓
$= 50/3$✓
$= 16{,}67\ \Omega$✓ (3)

1.2.4



(6)
[20]

QUESTION 2

2.1 2.1.1 True
2.1.2 False
2.1.3 True
2.1.4 False
2.1.5 True

(5 × 1) (5)

2.2
- Circuit components
- Supply voltage
- Output load
- Operation point of the output device
- Stray capacitance (5)

2.3 2.3.1 An amplifier is a device that produces an output signal with a waveform similar to that of the input signal, but the amplitude is amplified.

2.3.2 An oscillator is a device that produces an output signal of any desired frequency without any input signal.

(2 × 2) (4)

2.4 2.4.1
- Square-wave generator
- Voltage-to-frequency convertor
- Pulse synchronisation
- Clock for binary-logic signal
- Digital voltmeters and switched-mode power suppliers

(Any TWO)

2.4.2
- Produce rectangular waveforms
- Gating circuit
- Generate uniform pulse from variable-width input-pulse train

(Any TWO)

2.4.3
- Digital operations such as counters
- Storage of digital information
- Memory elements in shift registers and counters
- Frequency divider (Any TWO)

(3 × 2) (6)

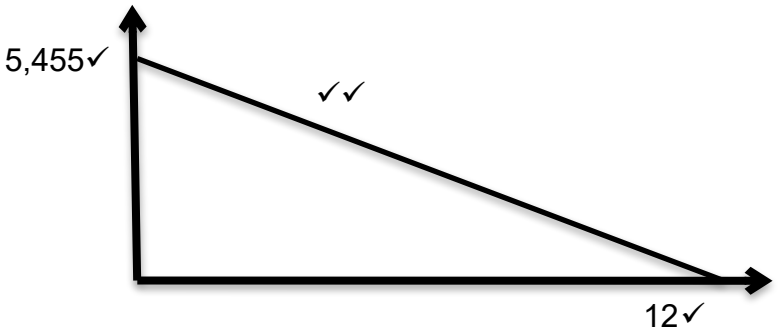
2.5 2.5.1 Switch-mode power supply
2.5.2 Uninterruptible power supply

(2 × 1) (2)

2.6 2.6.1 $$I_{CE}=\frac{V_{CC}}{R_C}=\frac{12}{2\ \ 200}=5{,}455\ \ mA$$ (3)

2.6.2 $$V_{CE}=V_{CC}=12\ \ V$$ (1)

2.6.3



(4)
[30]

QUESTION 3

3.1
- The strip of a copper track can loosen after components have been changed several times.
- When a copper track breaks, it may leave an invisible hairline of copper. This will result in an intermittent connection on the track, possibly causing short-circuiting.
- The relationship between the components and the circuit diagram may be confusing for beginners since the connecting tracks and where they are broken cannot be seen from the component side. This means that you have to turn the board over when fault-finding. (3)

3.2
- The energiser coil is faulty.
- The contact always remains in an open state.
- The contact always remains in closed state.
- The driver transistor to the relay is faulty. (4)

3.3 3.3.1 The replacement transistor should have a V_{CBO} rating equal to or higher than the original transistor.

3.3.2 The replacement transistor should have a V_{CEO} rating equal to or higher than the original.

3.3.3 The replacement transistor should have a V_{EBO} rating equal to or greater than the original.

3.3.4 The replacement transistor should be able to dissipate as much or more power as the original.

3.3.5 The replacement transistor should a gain equal to or better than the original transistor and should be as close to the original specification as possible.

(5 × 1) (5)

3.4
- It is a PNP transistor made from silicon material.
- It has a case outlet of type TO-92.
- The maximum collector-to-base voltage is 80 V.
- The maximum permissible collector current is 1 A.
- The lead-out connection is given by the number 24.
- The total power it can dissipate is 1 W. (6)

3.5 It is a device which electrically isolates one section of a circuit from another section. (2)

[20]

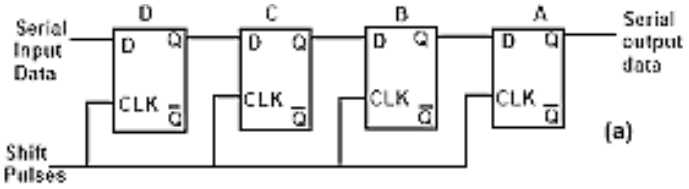
QUESTION 4

4.1.1 $0011\ 0100 = 34_{10}$ (1)

4.1.2 $0000_{BCD} = 0011_{xs-3}$ (1)

4.1.3 510 = 1011. (1)

4.2.1



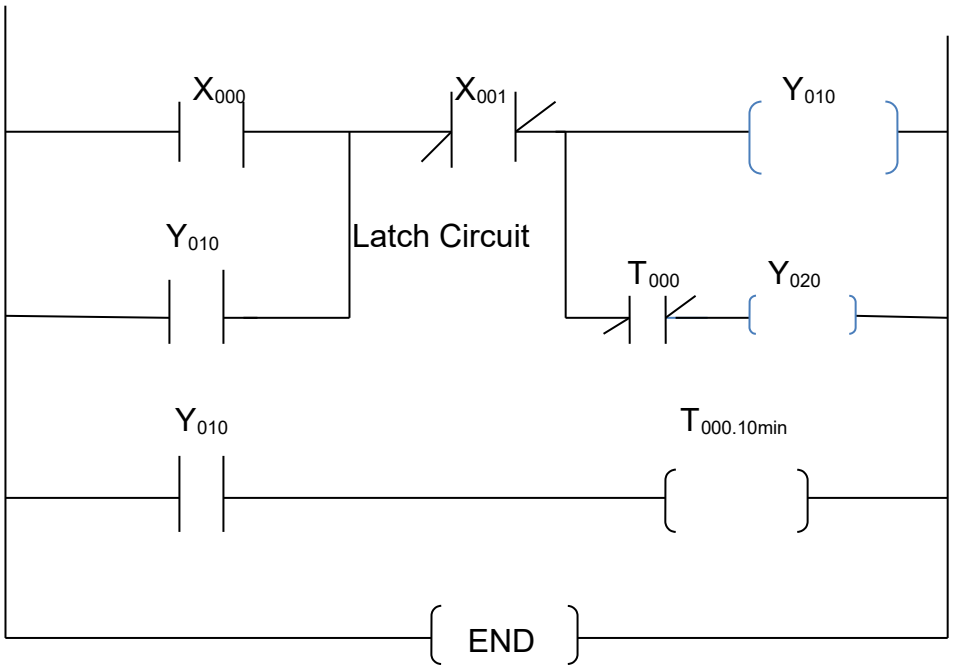
(3 marks for correct drawing and connections + 3 marks for labels) (6)

4.2.2 For every flip-flop in series as shown in 4.2.1, the data will be delayed by one clock pulse. ✓
Therefore, this shift register will delay the data by four clock pulses. ✓
Thus this shift register also acts as a temporary storage device✓ or it can act as a time delay device for the data, ✓ with the amount of time delay being controlled by the number of stages✓ in the register, 4, 8, 16 etc. or by varying the application of the clock pulses. ✓ (6)

[15]

QUESTION 5

5.1



(7)

5.2 In rung 1, when the START button is pressed, the motor (M1) is energised. √ via the normally closed contacts for STOP and T_1 timer√
Because motor M1 is running, its normally open contacts in the latch circuit below START become closed √ and the motor will remain energised even when the START switch is released.√
On the second rung normally open contacts for M1 are now closed thus energising timer T1.√
After 10 seconds timer T1 contacts on rung three close and the second motor (M2) is activated. √
Also after 10 seconds, the normally closed T1 contacts open and motor M1 stops running. . √
Motor M2 will continue running until the STOP button is depressed. √ (8)

[15]

TOTAL: 100

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